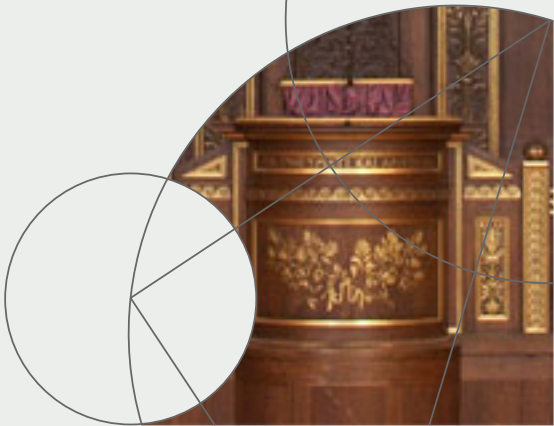
STEVENS INSTITUTE OF TECHNOLOGY

Data Acquisition and Processing I: Big Data

Rensheng Wang,
rwang1@stevens.edu
Dept. of Electrical and Computer Engineering
Stevens Institute of Technology



Final Project – Music Recommender

- You are given two sets of data:
  - Training data
  - Test data
- Training Data set contains the user rating scores for different Item IDs (i.e., Track ID, Album ID, Artist ID and Genre ID)
- Test Data set contains 6 Track IDs for each user.

☞ Task: You are asked to predict 3 "1"s and 3 "0"s for these 6 tracks based on which 3 track are more likely to be recommended.

Final Project: Test Data

- The snapshot of the test data: **testTrack_hierarchy.txt**

```
199810|208019|209288|None
199810|74139|277282|271146|113360|173467|173655|192976|146792|48505|133159
199810|9903|None|None|33722|123396|79926|73523
199810|242681|190640|244574|61215|17453|274088
199810|18515|146344|33168|19913|48505|154024
199810|105760|93458|11616|131552|173467|48505|133159
199812|276940|201356|163237|287681
199812|142408|112725|275191|158282|173467|242383|207648|48505|133159
199812|130023|226816|275191|158282|242383|207648|19913
199812|29189|104694|211701|158282|173467|81520|139095|48505|133159
199812|223706|101750|128069|158282|207648|279143
199812|211361|127464|19438|77904
```

- Each track belongs to a unique hierarchy structure, i.e.,:

  userID : trackID $\Rightarrow$ albumID $\Rightarrow$ artistID $\Rightarrow$ genreID

  For example, for the first two row records:

  199810 (userID) | 208019 (trackID) | 209288 (albumID) | None (artistID)

  199810 (userID) | 74139 (trackID) | 277282 (albumID) | 113360 (artistID) | 173467 (genreID1) | 173655 (genreID2) | 192976 (genreID3) | $\cdots$

Final Project: Test Data

- The snapshot of the training data: **trainIdx2_matrix.txt**

```
199808|248969|90
199808|2663|90
199808|28341|90
199808|42563|90
199808|59092|90
199808|64052|90
199808|69022|90
199808|77710|90
199808|79500|90
199808|82317|90
```

- Data format: userID | trackID | score

  For example, training user "199808" rated trackID "248969", with score "90".

Final Project: Check Training Records for Testing Users

- For the same user, we need to search all the related testing user item IDs in the corresponding training user records:

```
199810|208019|209288|None
199810|74139|277282|271146|113360|173467|173655|192976|146792|48505|133159
199810|9903|None|None|33722|123396|79926|73523
199810|242681|190640|244574|61215|17453|274088
199810|18515|146344|33168|19913|48505|154024
199810|105760|93458|11616|131552|173467|48505|133159
199812|276940|201356|163237|287681
199812|142408|112725|275191|158282|173467|242383|207648|48505|133159
199812|130023|226816|275191|158282|242383|207648|19913
199812|29189|104694|211701|158282|173467|81520|139095|48505|133159
199812|223706|101750|128069|158282|207648|279143
199812|211361|127464|19438|77904
```

test

```
199810|48050|70
199810|1589|50
199810|155767|70
199810|178994|50
199810|195282|50
199810|275191|50
199810|270655|50
199810|270460|30
199810|236943|50
199810|53052|70
```

training

Final Project: Check Training Records for Testing Users

- For the same user, we need to search all the related testing user item IDs in the corresponding training user records:

```
199810|208019|209288|None
199810|74139|277282|271146|113360|173467|173655|192976|146792|48505|133159
199810|9903|None|None|33722|123396|79926|73523
199810|242681|190640|244574|61215|17453|274088
199810|18515|146344|33168|19913|48505|154024
199810|105760|93458|11616|131552|173467|48505|133159
199812|276940|201356|163237|287681
199812|142408|112725|275191|158282|173467|242383|207648|48505|133159
199812|130023|226816|275191|158282|242383|207648|19913
199812|29189|104694|211701|158282|173467|81520|139095|48505|133159
199812|223706|101750|128069|158282|207648|279143
199812|211361|127464|19438|77904
```

test

```
199810|48050|70
199810|1589|50
199810|155767|70
199810|178994|50
199810|195282|50
199810|275191|50
199810|270655|50
199810|270460|30
199810|236943|50
199810|53052|70
```

training

- The sample python code at canvas:

  read_rating_v2.py